



**AMERICAN INTERNATIONAL UNIVERSITY–BANGLADESH
(AIUB)**

FACULTY OF SCIENCE & TECHNOLOGY

DEPARTMENT OF PHYSICS

PHYSICS LAB 5

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LAB REPORT ON

Determination of the temperature coefficient of resistance of the material of a wire

Supervised by,

DR. KAMRUN NAHAR MUKTA

Submitted By

S. N.	Name	ID	Contribution
1	Farhan Chowdhury	23-52314-2	Experimental Data, Discussion, Theory
2	Nayem Siddiki	23-52361-2	Theory, Discussion, Reference, Calculation
3	Fahim Al Noor	23-51977-2	Analysis, Discussion, Result, Theory
4	Amanur Rahman	23-52039-2	Procedure, Discussion, Reference, Theory
5	Mubashshirul Islam	23-52322-2	Apparatus, Result, Discussion

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1. Theory:

The determination of the temperature coefficient of resistance (TCR) of a wire involves measuring how the electrical resistance of the wire changes with temperature. This is crucial as different materials exhibit varying degrees of resistance changes with temperature fluctuations. To determine the TCR, the wire's initial resistance and temperature are measured at a starting point, usually at room temperature. Then, controlled heat is applied to the wire to raise its temperature while continuously measuring the resistance at specific temperature intervals. By calculating the ratio of the change in resistance to the change in temperature, the TCR can be obtained. A positive TCR value indicates an increase in resistance with temperature, while a negative value signifies a decrease in resistance with temperature. This experiment is essential for understanding and predicting the behavior of wires and other electronic components under varying temperature conditions, aiding in the design and functionality of electronic devices and system.

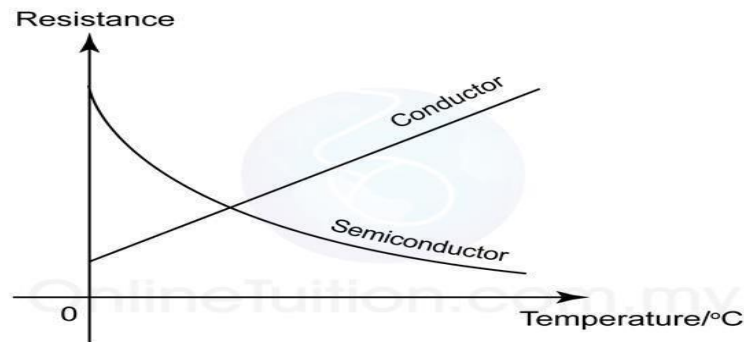


Figure 1: A graph that shows the changes in resistances of a conductor and a semiconductor with temperature. For conductor the resistance increases linearly and for semiconductor the resistance decreases exponentially with the temperature.

For a conductor the resistance increases with increase of temperature, as the figure 1 shows. If R_1 is the resistance of a conductor at temperature θ_1 and at a higher temperature θ_2 the resistance raises to R_2 , then we can write:

$$R_2 = R_1[1 + \alpha(\theta_2 - \theta_1)]$$

$$\text{or, } R_2 = R_1 + R_1\alpha(\theta_2 - \theta_1)$$

$$\text{or, } R_1\alpha(\theta_2 - \theta_1) = R_2 - R_1$$

$$\text{or, } \alpha = \frac{R_2 - R_1}{R_1(\theta_2 - \theta_1)}$$

$$\text{or, } \alpha = \frac{\Delta R}{R_1(\theta_2 - \theta_1)}$$

Here,

α = temperature coefficient of the material of the conductor.

ΔR = change in resistance due to the change in temperature $\Delta\theta$.

Taking $R_1 = 1 \text{ Ohms}$ & $\Delta\theta = 1^\circ\text{C}$ we get $\alpha = \Delta R$, thus we define the temperature coefficient of resistance of a substance as the change in resistance per unit resistance (per ohm) for unit change in temperature (per $^\circ\text{C}$). Also, we find the unit of α as per $^\circ\text{C}$.

For a conductor, if we know the resistances R_1 at a temperature θ_1 and R_2 at a higher temperature θ_2 , then we can calculate the temperature coefficient of resistance of its material by using the equation:

$$\alpha = \frac{R_2 - R_1}{R_1(\theta_2 - \theta_1)}$$

The unknown resistance can be determined by a meter bridge that works with the Wheatstone bridge principle.

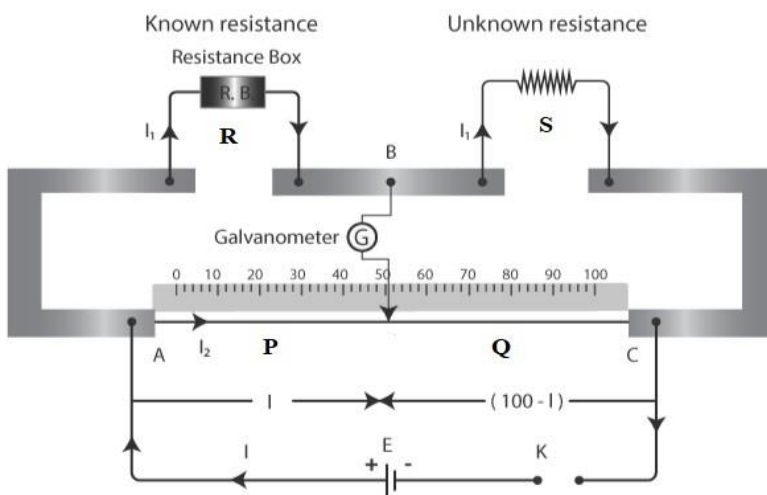


Figure 2: A meter bridge used to determine an unknown resistance by **using** the Wheatstone bridge principle.

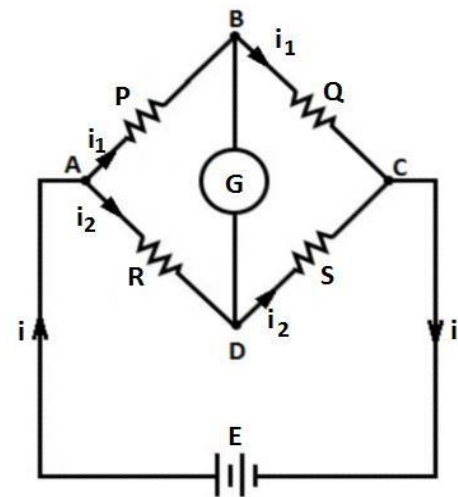


Figure 3: The Wheatstone Bridge

In the meter bridge circuit as in figure 3, for the null deflection in the galvanometer we get,

$$\frac{P}{Q} = \frac{R}{X}$$

$$\Rightarrow X = \frac{Q}{P} \times R = \frac{\rho(100 - l)/A}{\rho l/A} \times R = \frac{100 - l}{l} \times R,$$

According to above equation, if we know the length of the balance point, l and the resistance, R we can determine the unknown resistance S .

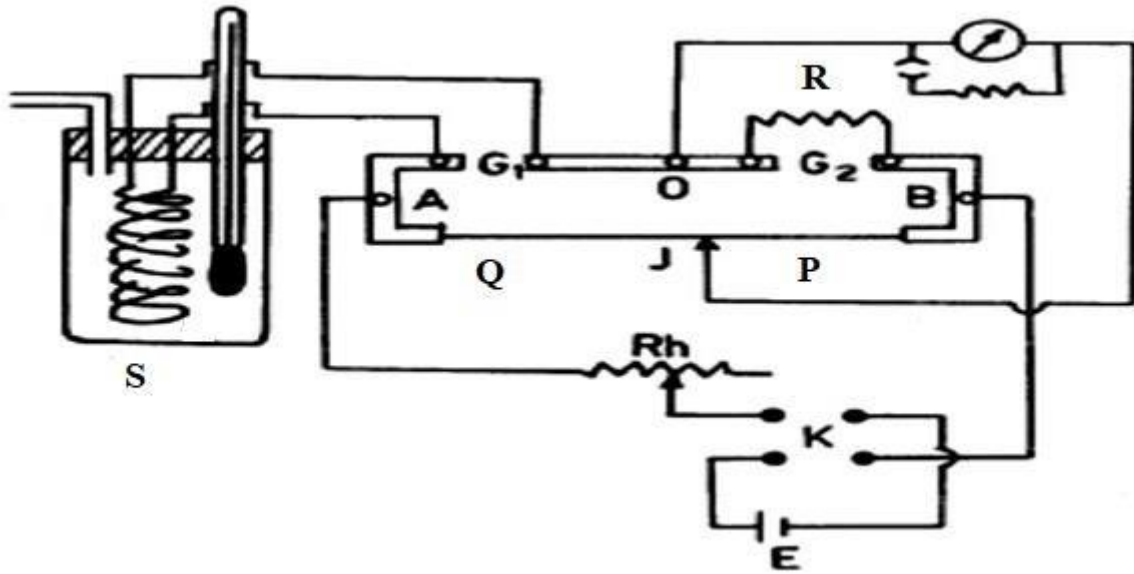


Figure 4: Circuit Diagram for the experiment

2. Apparatus:

1. Power Supply
2. Meter Bridge
3. Galvanometer
4. Jockey, Resistance Box
5. Coil of conducting wire
6. Thermometer
7. Water, Electric heater
8. Commutator Beaker

3. Procedure

The figure 4 apparatus was utilized as follows:

- i. Initially, the coil of wire was connected to the left gap (G_1) and placed inside the beaker at room temperature. θ_1 from the thermometer was recorded. Subsequently, the resistance box was linked to the right gap (G_2) of the meter bridge. The flow of electricity through the circuit was altered by manipulating the knobs in different positions of the commutator. Various resistances (e.g., 5, 7, 9 Ohms) from the resistance box were employed, and the equilibrium positions on the meter bridge were identified. The measurements for the reverse current were also recorded.

- ii. The positions of the resistance box and the coil of wire were interchanged on the meter bridge gaps. Direct and reverse current readings for identical resistances (e.g., 5, 7, 9 Ohms) were also documented.
- iii. The temperature within the beaker was elevated by introducing hot water. The elevated temperature (θ_2) was noted using the thermometer. The equilibrium points for the same resistances (e.g., 5, 7, 9 Ohms) were once more determined for both direct and reverse currents, considering connections of the coil of wire to the right and left gaps of the meter bridge.
- iv. The value of the unknown resistance was determined for both left and right gaps. Additionally, the temperature coefficient (α) was calculated

3. Experiment Data:

Temperature $^{\circ}\text{C}$	Known Resistance, R (Ohms)	Position of Unknown Resistance, S	Position of balance point, l (cm)		Mean of l (cm)	$100 - l$ (cm)	S (Ohms)	Mean of S (Ohms)
			Direct current	Reverse current				
24 = θ_1	5	Left	58.2	59.1	58.65	41.35	3.53	$X_1 = 6.35 \Omega$
		Right	49.5	50.5	50	50	5	
	7	Left	52.5	51.9	52.2	47.8	6.41	
		Right	54.5	51.8	53.15	46.85	6.17	
	9	Left	47.3	45.5	46.4	53.6	10.39	
		Right	59.3	55.9	57.6	42.4	6.63	
80 = θ_2	5	Left	58.1	56.1	57.1	42.9	3.76	$X_2 = 6.84 \Omega$
		Right	44.5	40.2	42.35	57.65	6.81	
	7	Left	56.9	50.7	53.8	46.2	6.01	
		Right	46.7	49.6	48.15	51.85	7.54	
	9	Left	54.3	45.4	49.85	50.15	9.05	
		Right	52.6	54.1	53.35	46.65	7.87	

Table – 1: Readings of balance points on Meter Bridge.

4. Calculation:

$$\begin{aligned}
 \text{Temperature coefficient of resistance, } \alpha &= \frac{X_2 - X_1}{X_1(\theta_2 - \theta_1)} \\
 &= \frac{6.35 - 6.35}{6.35(80 - 24)} \\
 &= 1.378 \times 10^{-3} \text{ C}^{-1}
 \end{aligned}$$

5. Result:

Temperature	Resistance (ohm)	Temperature coefficient of resistance ($^{\circ}\text{C}$)
$\theta_1 = 24$	$X_1 = 6.36$	$1.378 \times 10^{-3} \text{ C}^{-1}$
$\theta_2 = 80$	$X_2 = 6.67$	

7. Discussion:

The experiment aimed to determine the temperature coefficient of resistance (TCR) of a wire's material. The wire's resistance was measured at different temperatures using a multimeter. The temperature was controlled and varied systematically during the experiment. By plotting the resistance versus temperature data and calculating the slope of the graph, the TCR was obtained. This coefficient indicates how much the wire's resistance changes with temperature, providing valuable information for practical applications and understanding the wire's behavior under different conditions.

8. References:

- Fundamental of Physics (10th Edition): Resistance and resistivity (Chapter 26, page-755)
- Video Links:
 - Temperature coefficient: <https://www.youtube.com/watch?Tgm 0f2mOs>
 - Meter Bridge: <https://www.youtube.com/watch?venqx8vIdHVKQ>